Stage setting: Hubble anisotropy from an off-center CoE — derivation, Migkas ladder, CMB toy, and GR next

Chris
hedge / cosmology notes

2026-09-20

Abstract

This note stages the Newtonian Sedov–Taylor (ST) + center-of-expansion (CoE) program for an off-center observer. Homologous expansion about a real CoE yields an isotropic apparent Hubble law; anisotropy enters only through Gauss gravity with radially decreasing density. Against the Migkas et al. (2020) $A(\ell)$ pattern (digitized Fig. 6), first-principles ST+Gauss geometry alone reaches $R^2 \approx 0.543$. A diagnostic Amp$\times$ sin phase rotation raises $R^2_A \approx 0.746$; a multiplicative 2ℓ residual layer tuned on A reaches $R^2_A \approx 0.875$. A concentric blackbody-shell toy for the CMB dipole requires $R/D \sim 203$ to match Planck's 3362 μK amplitude under $\delta T/T = (1/4)\,\delta F/F$. Numbers below are taken from on-disk fits and files; Amp$\times$ sin and 2ℓ are phenomenological. Much work remains for a true best fit. Relativity (LTB or cousin) is required as the next chapter.

Contents

1 Basic Hubble vector derivation **3**

2 A versus angle model **3**
2.1 Geometry 3
2.2 ST blast + Gauss gravity with decreasing $\rho(r)$ 4

3 Observable from Migkas **5**

4 Residuals: model A versus Migkas (before rotation) **5**

5 Rotation (Amp$\times$ sin) **5**

6 Residuals after rotation **5**

7 2ℓ residual **6**

8 Migkas versus model scatter **6**

9 CMB dipole from shifting blackbody concentric shells **6**

10 2D maps (Mollweide) **7**

11 Stage setting / work remaining for a true best fit **8**

12 Relativity required **9**

1 Basic Hubble vector derivation

Place a real geometric center of expansion (CoE) at the origin. The observer sits at position $\vec{D}$ with $D = \|\vec{D}\|$. A source sits at $\vec{r}$ (CoE-radius $r = \|\vec{r}\|$). The observer–source separation is

$$\vec{d} = \vec{r} - \vec{D}, \qquad d = \|\vec{d}\|, \qquad \hat{n} = \vec{d}/d. \tag{1}$$

Assume homologous expansion about the CoE,

$$\vec{v}(\vec{r}) = H\,\vec{r}, \tag{2}$$

with the same H for every tracer (Sedov–Taylor interior working assumption). Then the observer and source velocities in the CoE frame are $\vec{v}_O = H\vec{D}$ and $\vec{v}_S = H\vec{r}$, so the relative velocity is

$$\vec{v}_{\rm rel} = \vec{v}_S - \vec{v}_O = H(\vec{r} - \vec{D}) = H\vec{d}. \tag{3}$$

The line-of-sight recession speed and apparent Hubble parameter are

$$v_{\rm los} = \vec{v}_{\rm rel} \cdot \hat{n} = H\,d, \qquad H_{\rm obs} \equiv \frac{v_{\rm los}}{d} = H. \tag{4}$$

The CoE cancels: every off-center observer measuring relative velocities sees the same isotropic H. Anisotropy is *not* a kinematic consequence of being off-center under pure homology. Something must break homology—here, gravity from a centrally concentrated, radially decreasing density (Sec. 2).

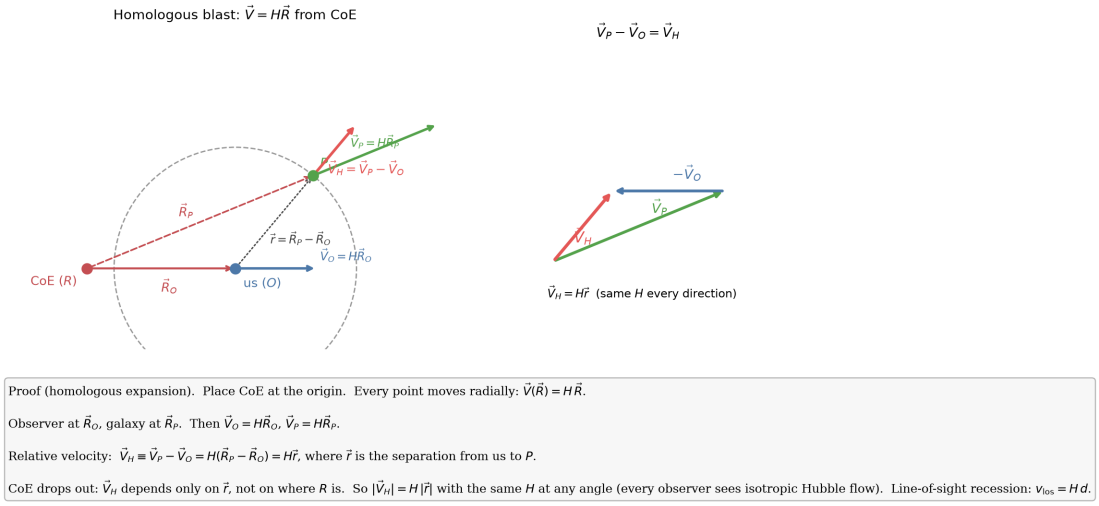


Figure 1: Homologous expansion about a CoE: absolute velocities point away from the center, but relative velocity between observer and source is $\vec{v}_{\rm rel} = H\vec{d}$, yielding isotropic $H_{\rm obs} = H$.

2 A versus angle model

2.1 Geometry

Observer at distance D from the CoE; sources sampled on an observer-centered circle of depth d. Earth-centric angle φ is measured from the preferred axis $\hat{D} = \vec{D}/D$. Scale length of the mass profile is L; dimensionless ratios of interest are d/D, D/L, and mass ratio M/M_0.

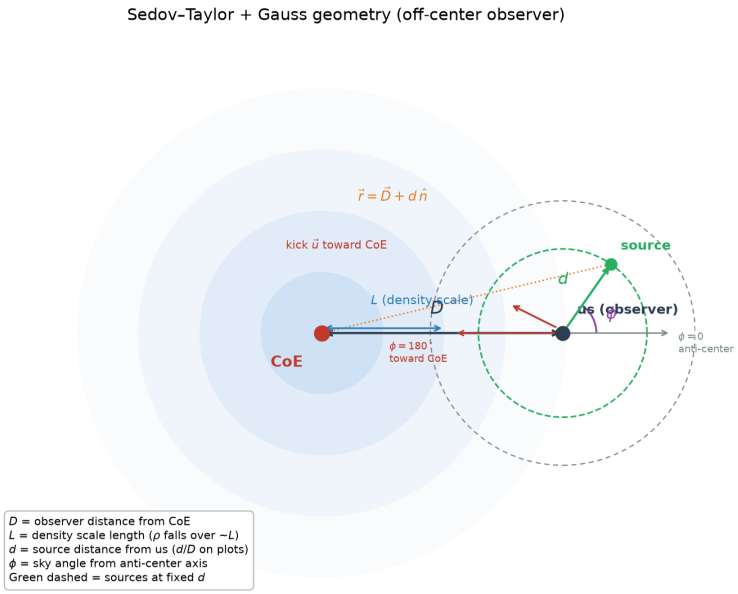


Figure 2: Off-center observer geometry: CoE, observer at D, source distance d, angle φ from the preferred axis.

2.2 ST blast + Gauss gravity with decreasing $\rho(r)$

Classical ST blast front: $R_{\rm ST}(t) \propto t^{2/5}$, $H = \dot{R}_{\rm ST}/R_{\rm ST} = 2/(5t)$. Background velocities remain homologous (Eq. 2). Gravity is *not* a uniform ball ($g \propto -r$). For spherical $\rho(r)$ decreasing outward, Gauss / shell theorem gives

$$\vec{g}(r) = -\frac{GM(<r)}{r^2}\,\hat{r}. \tag{5}$$

A dimensionless gravity strength relative to expansion is

$$\kappa \equiv \frac{GM(<D)}{D^3 H^2}. \tag{6}$$

A natural order-of-magnitude target is $\kappa \sim \Omega_m/2 \approx 0.15$ if one identifies enclosed mass with a cosmic mean; fitted working values in this program are larger (Sec. 3–5).

Peculiar velocities from gravity break homology. In the sheet AF convention used here,

$$A \propto \frac{H_{\rm obs}}{H} \tag{7}$$

(mean-normalized when compared to Migkas). Ideal continuous integration of gravity along expansion history rescales amplitude ($\alpha \to \alpha_{\rm eff}$) but does not flip the angular shape of $A(\varphi)$; expand versus collapse does.

Toy-universe ranges from fits (not informal guesses). On-disk continuous ideal fit against digitized Migkas A (`ideal_fit_M_d_D_L_migkas.npz`): $d/D \approx 0.76$, $D/L \approx 7.62$, $\kappa \approx 0.458$, $R^2 \approx 0.78$. Discrete geometry-grid best (before Amp$\times$sin in the staged ladder): $M/M_0 = 3$, $d/D = 0.5$, $L \approx 0.169$, $D/L \approx 5.9$, $\kappa \sim 0.7$, geometry-alone $R^2 \approx 0.543$. These fitted ranges—not informal 3–4r guesses—are the working toy universe for this stage-setting note.

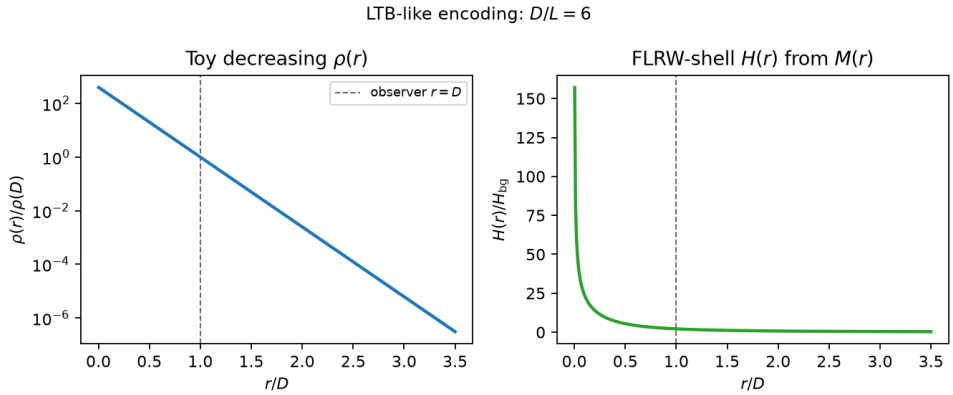


Figure 3: Toy density and Hubble profile versus CoE radius (from the GR Ch. 2 draft figures). Decreasing $\rho(r)$ is the ingredient that makes Gauss gravity anisotropic for an off-center observer.

3 Observable from Migkas

Migkas et al. (2020) report directional variation in the X-ray cluster L_X–T normalization A (and related B) versus Galactic longitude (their Fig. 6). Digitized points are from `migkas/fig6_digitized.csv` (copied into this paper's `figures/`). Those data remain contested (calibration, absorption, bulk flow, scatter); they are used here as the empirical angular target, not as settled cosmology.

Geometry alone (grid best cited above) gives $R^2 \approx 0.543$ on A. That is the first-principles rung of the ladder; subsequent Amp$\times$ sin and 2ℓ layers are diagnostic, not derived ST+Gauss.

4 Residuals: model A versus Migkas (before rotation)

Figure 6 shows Migkas A minus the geometry (sheet AF) template after a least-squares affine match in amplitude/offset. Systematic angular structure remains—the motivation for the Amp$\times$ sin diagnostic in Sec. 5.

5 Rotation (Amp$\times$ sin)

A one-parameter phase alignment on the longitude circle,

$$A_{\rm rot}(\ell) = A_{\rm geom}(\ell)\,\big[1 + \mathrm{Amp}\cdot\sin(\ell+\phi)\big] \tag{8}$$

(with Amp and ϕ fitted; sheet convention), raises the A match substantially. Best Amp$\times$ sin values used here (from `R2_best_A_B_Amp_phi_summary.csv` / `rotation_Amp_phi_R2_ranked.csv`):

$$\mathrm{Amp} = -0.06, \qquad \phi \approx 11.46^\circ\ (0.20\,\mathrm{rad}), \qquad R^2_A \approx 0.746. \tag{9}$$

Shared opposite amplitude $\mathrm{Amp}_B = -\mathrm{Amp}_A$ gives $R^2_B \approx 0.79$ at the same ϕ. Geometry alone was $R^2_A \approx 0.543$; Amp$\times$ sin is a diagnostic phase knob, not a first-principles ST prediction.

6 Residuals after rotation

After Amp$\times$ sin, residuals shrink but retain structure (Fig. 11). The companion panel with a sin / cos guide (Fig. 12) motivates the 2ℓ layer of Sec. 7.

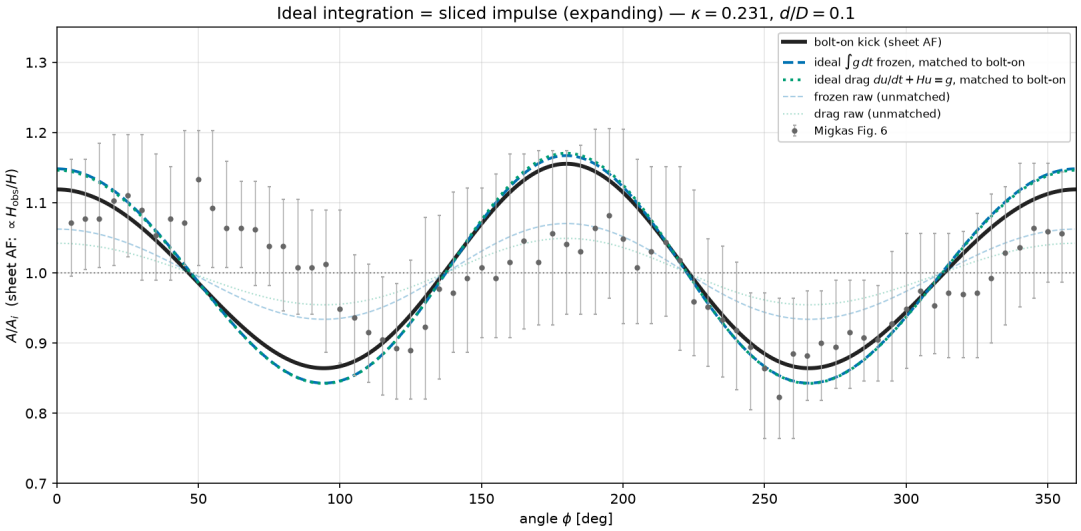


Figure 4: Ideal-integration ST+Gauss A versus angle under sheet AF ($A \propto H_{\rm obs}/H$), compared against the Migkas longitude pattern after axis phasing.

7 2ℓ residual

A multiplicative residual layer tuned on A,

$$A_{2\ell}(\ell) = A_{\rm rot}(\ell)\,\big[1 + \varepsilon\cos(2\ell + \psi)\big], \tag{10}$$

with on-disk best values from `best_fit_2ell_multiplier.npz`:

$$\varepsilon \approx -0.043, \qquad \psi \approx 176^\circ, \qquad R_A^2 \approx 0.875 \tag{11}$$

(RMSE $0.038 \to 0.029$ on that chain). This is phenomenological: it absorbs leftover quadrupole-like structure after geometry + Amp$\times$ sin.

Honest summary of the A ladder:

Stage	Key parameters	R_A^2
Geometry alone	$M/M_0 = 3$, $d/D = 0.5$, $L \approx 0.169$, $\kappa \sim 0.7$	≈ 0.543
+ Amp$\times$ sin	Amp$= -0.06$, $\phi \approx 11.46^\circ$	≈ 0.746
+ 2ℓ (A-tuned)	$\varepsilon \approx -0.043$, $\psi \approx 176^\circ$	≈ 0.875

8 Migkas versus model scatter

Scatter plots make the R^2 meaning concrete: points lining up on $y = x$ after the Amp$\times$ sin (and related) adjustments.

9 CMB dipole from shifting blackbody concentric shells

Fixsen monopole $T_0 \approx 2.72548\,\mathrm{K}$. Planck 2018 dipole amplitude $A_{\rm dip} = 3362.08\,\mu\mathrm{K}$, so

$$\left(\frac{\delta T}{T}\right)_{\rm amp} \approx \frac{3362.08 \times 10^{-6}}{2.72548} \approx 1.23 \times 10^{-3}. \tag{12}$$

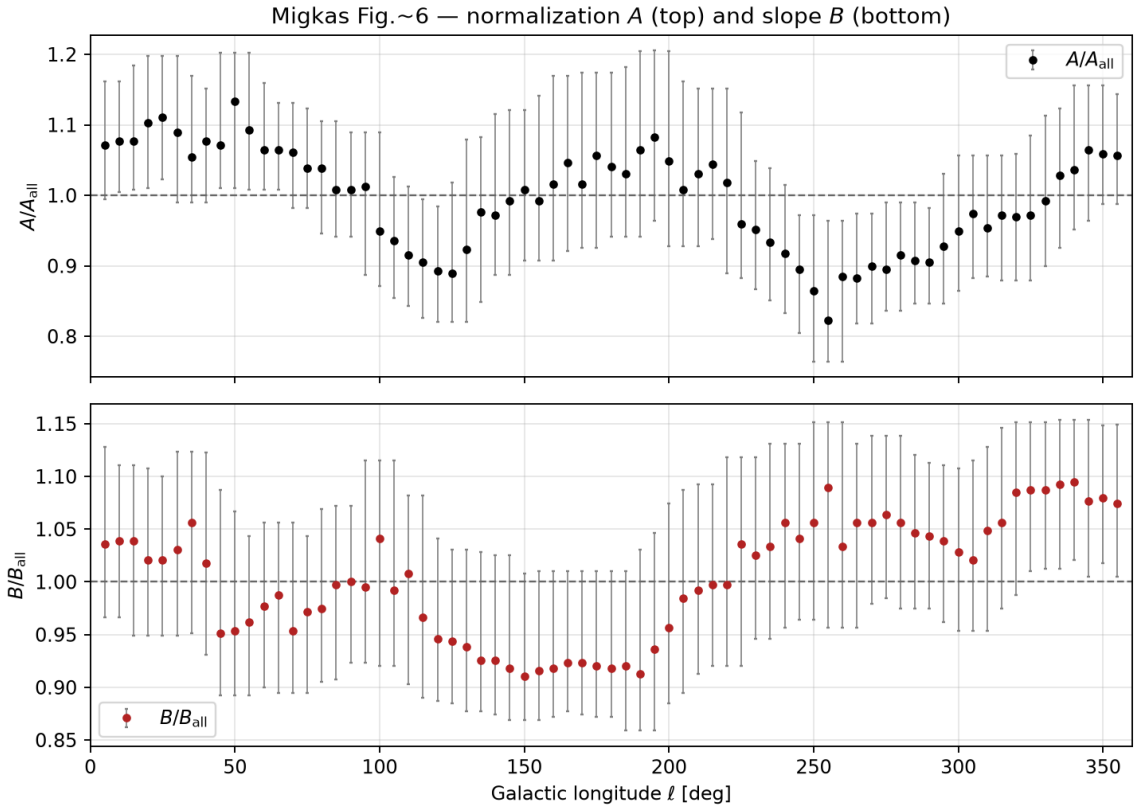


Figure 5: Digitized Migkas et al. (2020) Fig. 6: A and B versus Galactic longitude, with model overlays from the ST+CoE ladder where available.

Hot–cold peak-to-peak is $\approx 6.72\,\mathrm{mK}$. For blackbody intensity,

$$\frac{\delta T}{T} = \frac{1}{4}\frac{\delta F}{F}, \tag{13}$$

so matching the dipole amplitude needs $(\delta F/F)_{\mathrm{amp}} \approx 4.93 \times 10^{-3}$.

Concentric-shell toy. Outer-shell heating offset relative to an inner shell can source a sky dipole for an off-center observer. Scenarios 0, 2, and 4 are retained; Scenario 1 is dropped for producing an unacceptably large shift. In the thin-shell picture, $T_{\mathrm{in}} \approx T_{\mathrm{out}} \approx T_0$; a cavity (Kirchhoff) would kill the dipole. Geometric flux contrast scaling as $\delta F/F \sim D/R$ implies

$$\frac{R}{D} \approx 203 \tag{14}$$

to reach $(\delta F/F) \approx 4.93 \times 10^{-3}$. That is a kinematic/flux toy, not a radiative-transfer solution.

10 2D maps (Mollweide)

Full-sky products in this program remain provisional. Figure 22 is the existing `sky_mollweide.png`: useful as a stage-setting sky sketch, not a polished publication map. Figure 23 is a cleaner Amp × sin

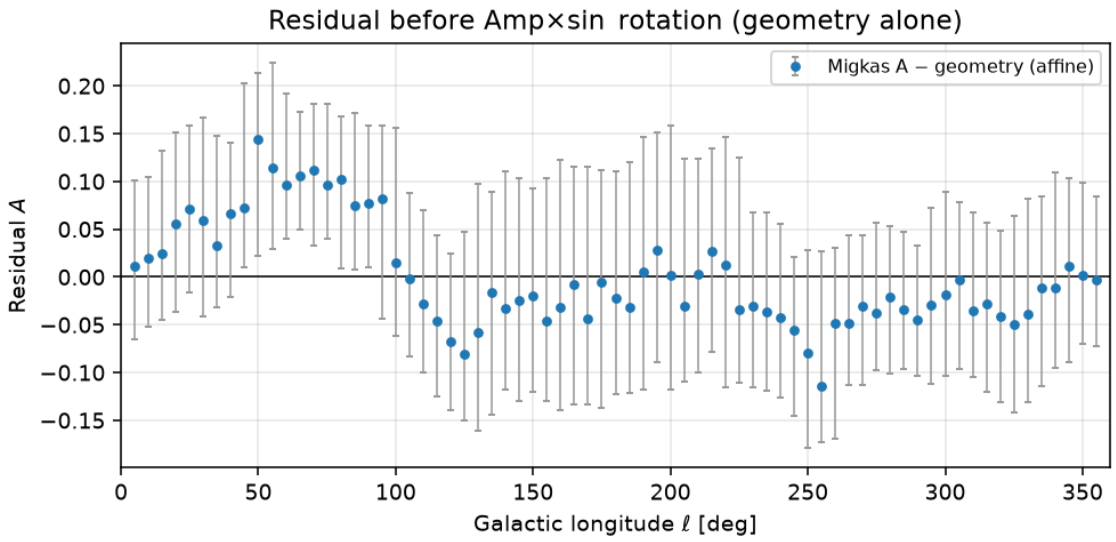


Figure 6: Residual before Amp× sin rotation: digitized Migkas A minus geometry (AF) template with affine match. Error bars from the digitized Fig. 6 uncertainties.

dipole-proxy Mollweide locked to the CoE axis inferred from Amp× sin + Planck $|b|$:

$$\begin{aligned} \text{CoE/cold} &\approx (258.5^\circ, +48.3^\circ), \\ \text{anti-CoE/hot} &\approx (78.5^\circ, -48.3^\circ), \end{aligned} \tag{15}$$

with the usual hot/cold flip caveat relative to a pure Planck kinematic dipole reading. This is *not* a full ST+Gauss projected A map—only an axis lock for orientation.

11 Stage setting / work remaining for a true best fit

This note stages the Newtonian ladder. It is not a finished cosmological fit. Much work remains:

- **Joint A+B likelihood**, not A-first then associated B.
- **3D axis** (longitude + latitude) instead of a longitude-circle phase only; lock or float against Planck $|b|$ with explicit hot/cold convention.
- **Time-integrated gravity** end-to-end in the likelihood (not only amplitude rescaling after a frozen snapshot).
- **Selection and survey geometry** for Migkas-like cluster samples (mask, redshift slice, Malmquist / absorption systematics).
- **Replace Amp× sin and 2ℓ** with physics-derived quadrupole / line-of-sight depth effects, or retire them and report geometry alone.
- **Ideal continuous vs discrete grid**: reconcile $(d/D, D/L, \kappa)$ posteriors with shared conventions for A normalization.
- **CMB**: move from flux-shell toys to last-scattering and ISW in a metric model; keep Alnes/Kodama-class bounds as adversaries.

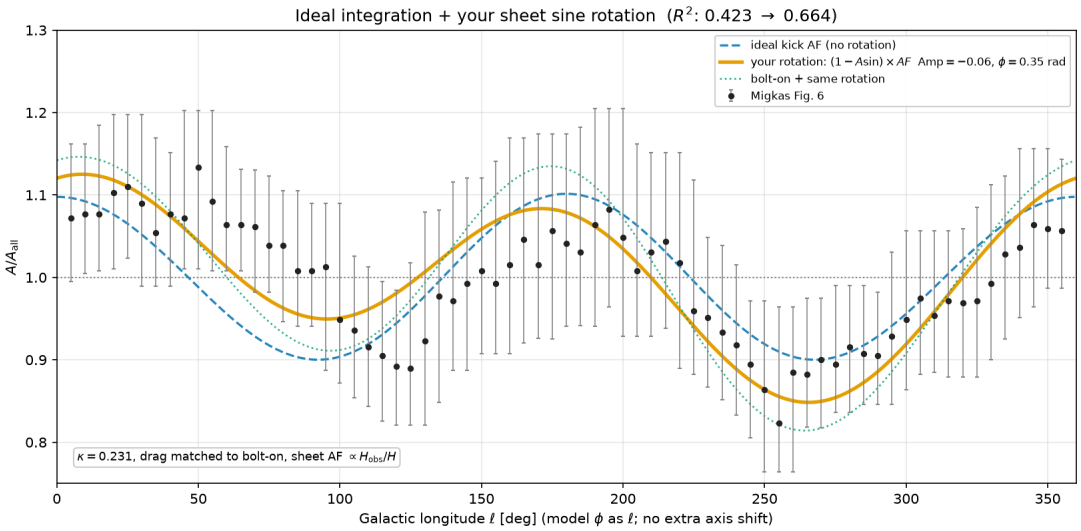


Figure 7: Ideal geometry $A(\ell)$ plus Amp$\times$ sin rotation versus Migkas points.

- **Honest error model**: digitize uncertainties, covariance across ℓ bins, and look-elsewhere for Amp–ϕ–ε scans.

12 Relativity required

Newtonian ST+Gauss is scaffolding. A real CoE beyond the horizon, null geodesics, and the CMB demand a relativistic inhomogeneous model—LTB or a close cousin—as the next chapter. The existing draft `papers/gr_ch2_ltb_2026-09-20/main.pdf` sketches that program. Priority order (Chris): get the Hubble / Migkas side right first; park Alnes/Kodama-class CMB dipole bounds as the principal adversary rather than pretending the Newtonian shell toy closes the case. Until a metric model carries both $H_{\rm obs}(\hat{n})$ and the CMB, claims of a "best fit universe" remain premature.

Acknowledgments

Conversations and genesis: Wolfgang Rindler; Luke Ackerman.

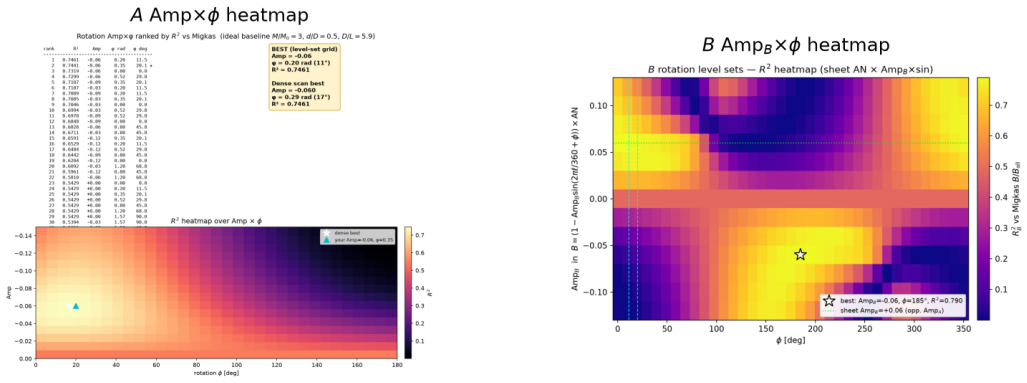
References

[1] K. Migkas, G. Schellenberger, T. H. Reiprich, F. Pacaud, M. E. Ramos-Ceja, and L. Lovisari, "Probing cosmic isotropy with a new X-ray galaxy cluster sample through the L_X–T scaling relation," *Astron. Astrophys.* **636**, A15 (2020).

[2] Planck Collaboration, "Planck 2018 results. I. Overview and the cosmological legacy of Planck," *Astron. Astrophys.* **641**, A1 (2020).

[3] D. J. Fixsen, "The Temperature of the Cosmic Microwave Background," *Astrophys. J.* **707**, 916 (2009).

[4] H. Alnes and M. Amarzguioui, "CMB anisotropies seen by an off-center observer in a spherically symmetric inhomogeneous universe," *Phys. Rev. D* **74**, 103520 (2006).



Best R^2 — Amp×sin rotation level sets (sheet AF / AN)

Model	R_A^2	R_B^2	Parameters
Geometry / AN alone (no Amp×sin)	0.543	0.483	—
Best Amp×φ (solo scan)	0.746	0.790	A: Amp=-0.06, φ=11.5° B: Amp_B=-0.06, φ=184.9°
Shared-opp at A best φ (Amp_B=−Amp_A)	0.746	0.790	Amp_A=-0.06, Amp_B=+0.06, φ≈11.5°
A + multiplicative 2ℓ (A-tuned)	0.875	— (hurts B if re	ε≈−0.043, ψ≈176°

Figure 8: R^2 tables / Amp–ϕ heatmaps for A and B under the Amp$\times$ sin scan.

[5] H. Kodama, K. Saito, and A. Ishibashi, "Analytic solutions for the constrained cosmological perturbation equations in the matter-dominated era and CMB anisotropies," related LTB / off-center CMB literature as cited in program digests (2010).

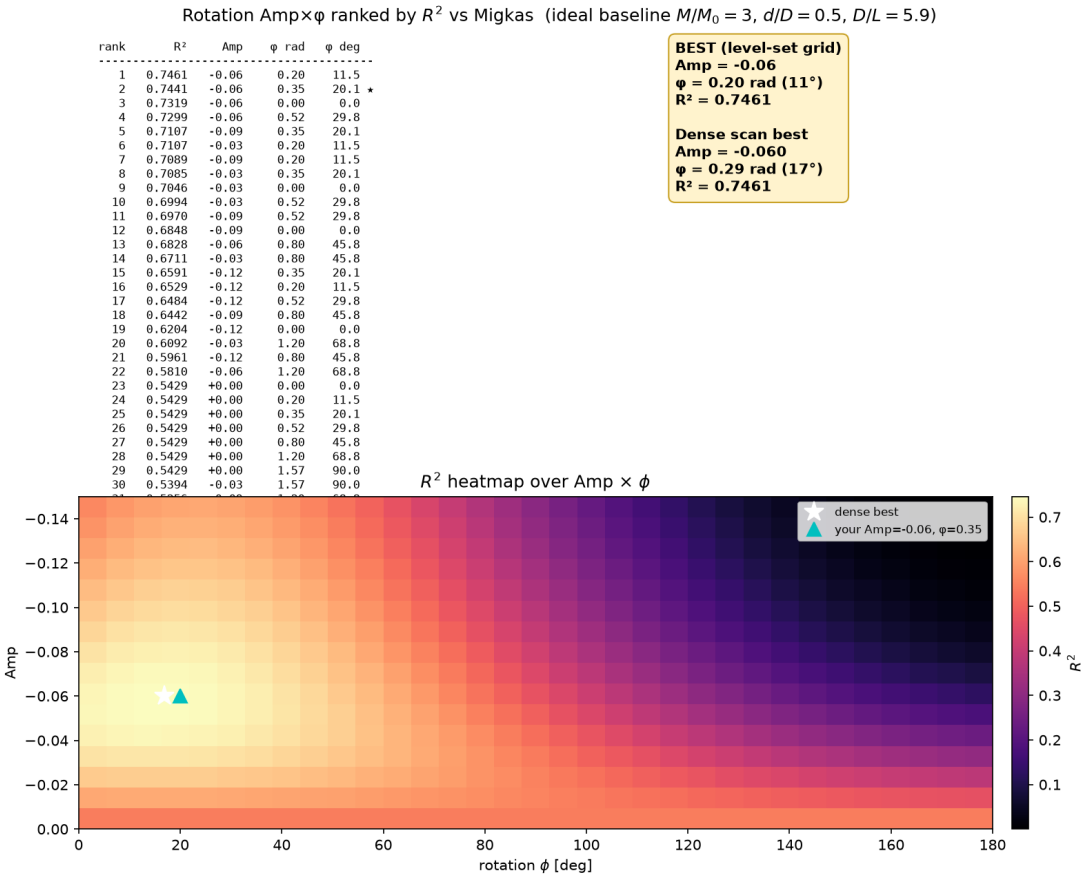


Figure 9: Ranked Amp–ϕ scan for R_A^2 (peak near Amp$= -0.06$, $\phi \approx 11.5°$).

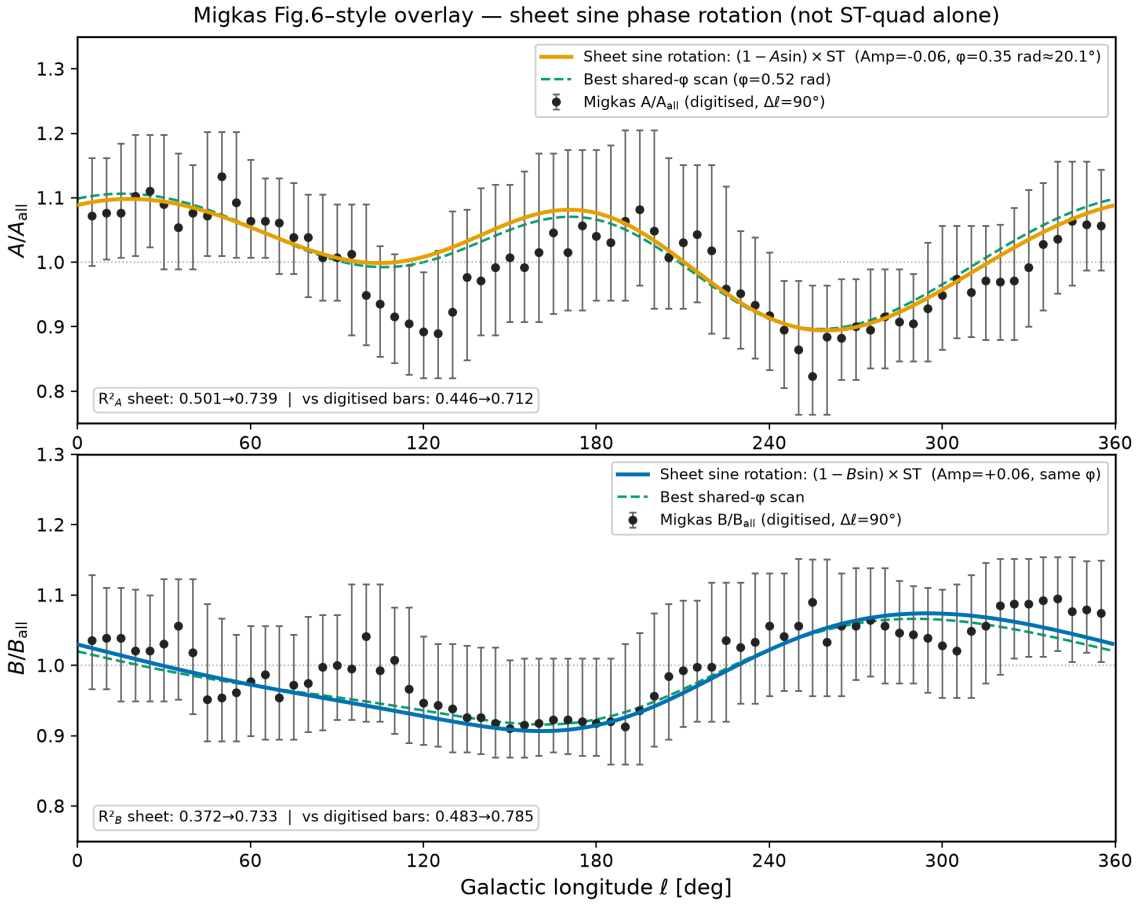


Figure 10: Migkas bars with Amp× sin rotation overlay (sheet-style diagnostic).

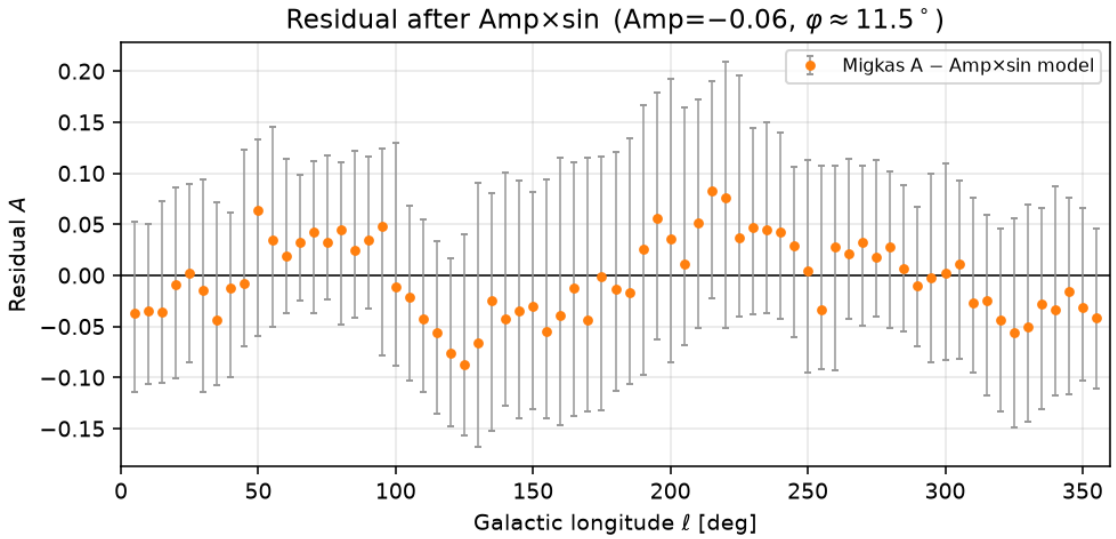


Figure 11: Residual after Amp× sin (Amp= −0.06, $\phi \approx 11.5°$): Migkas A minus rotated model (on-disk Amp× sin chain).

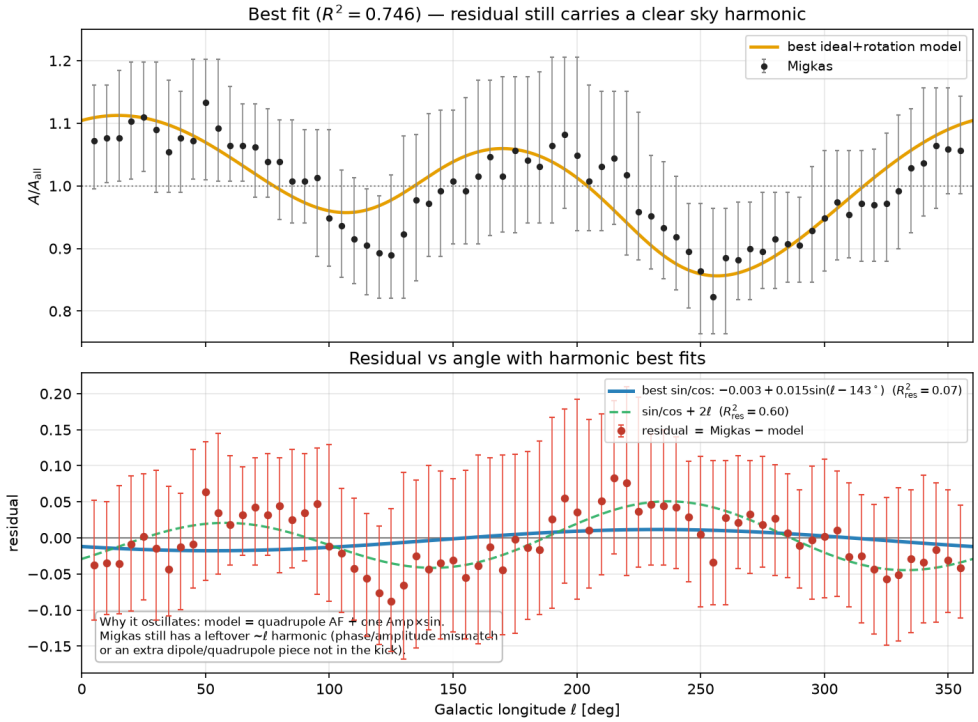


Figure 12: Post-rotation residual versus ℓ with sin / cos guide curves (existing fit product; used as input to the 2ℓ multiplier).

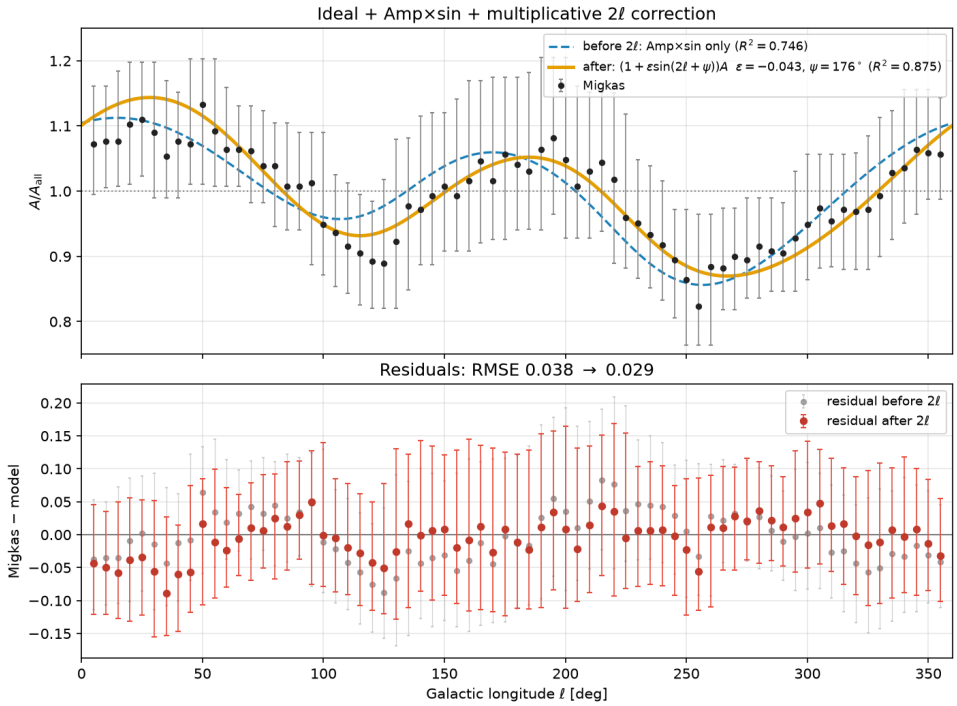


Figure 13: Best-fit Amp× sin model with multiplicative 2ℓ layer on A ($R_A^2 \approx 0.875$).

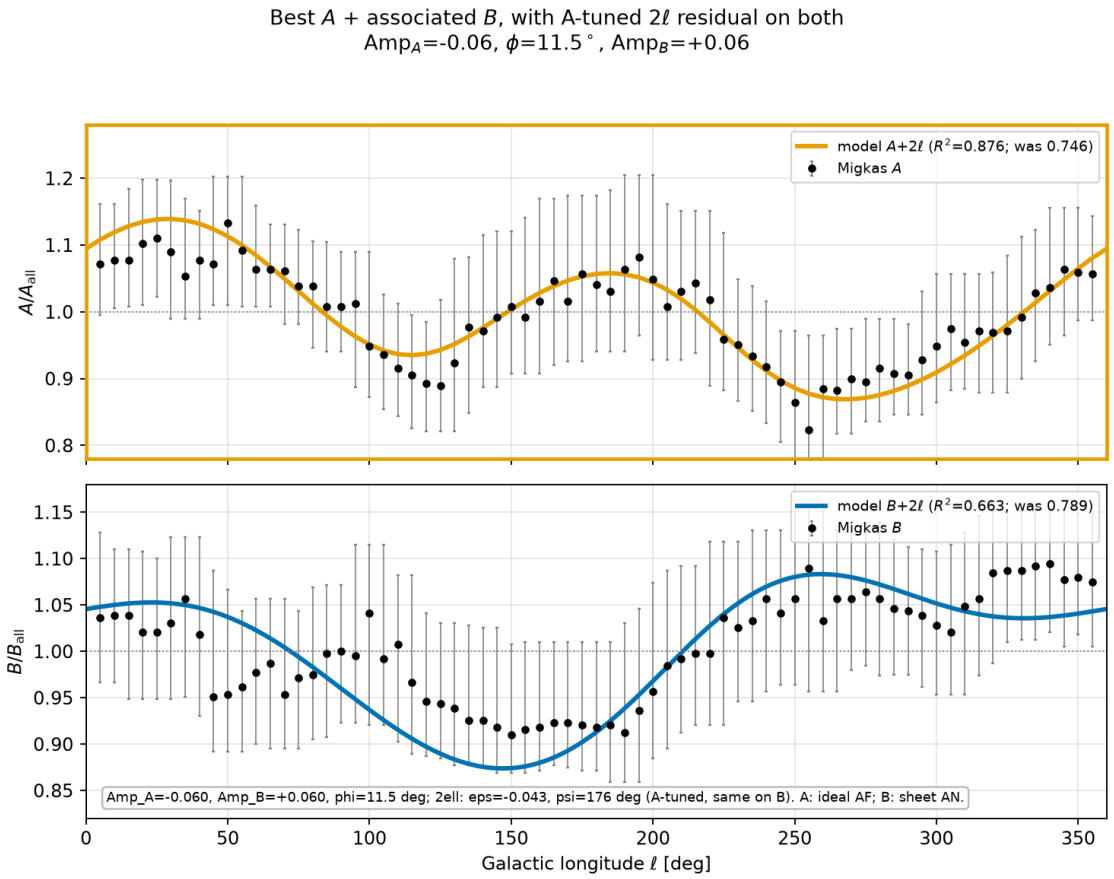


Figure 14: A-tuned 2ℓ fit with associated B shown. B does not improve the same way when the 2ℓ layer is tuned on A.

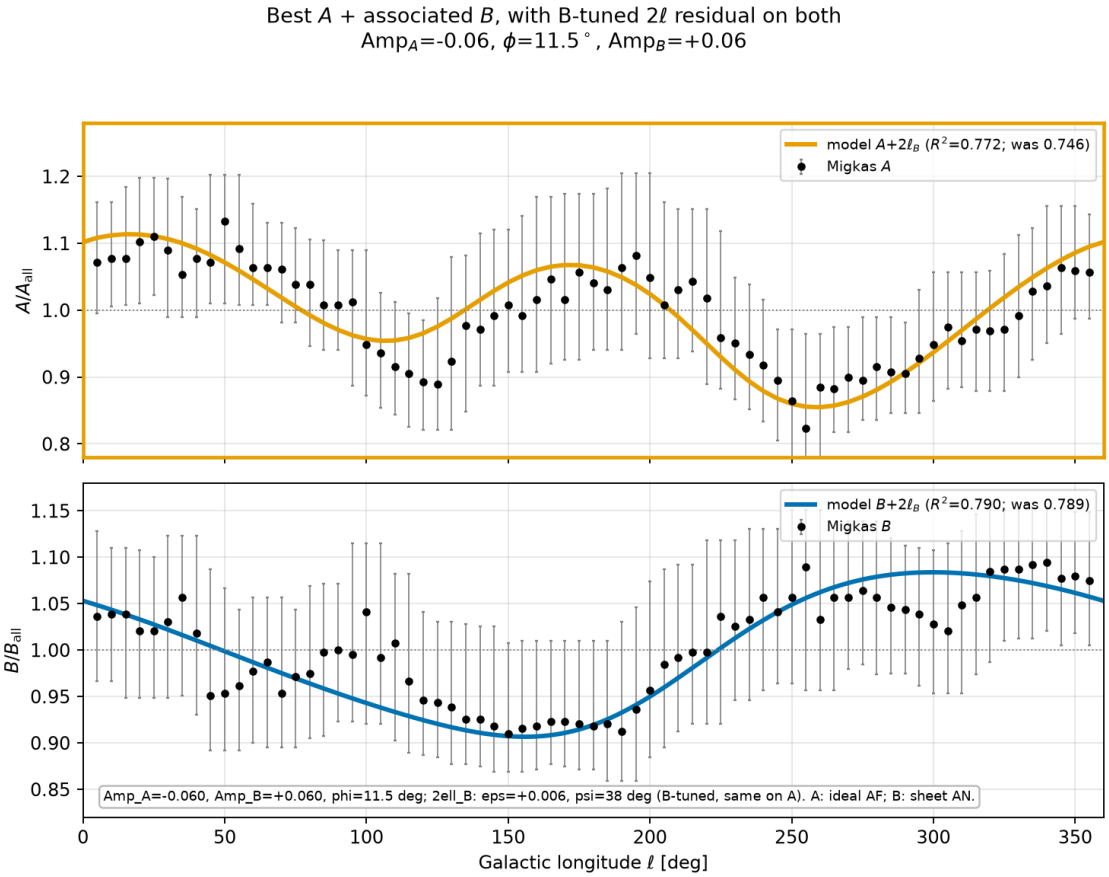


Figure 15: Companion: 2ℓ chain when tuned from B (for contrast; A is the primary Migkas target in this note).

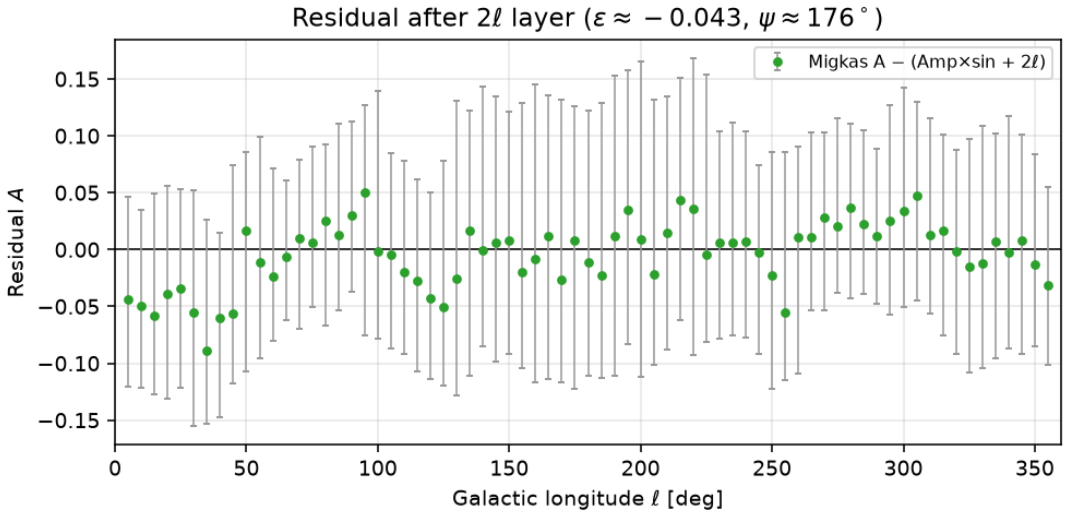


Figure 16: Residual after the A-tuned 2ℓ layer ($R_A^2 \approx 0.875$ on the same chain).

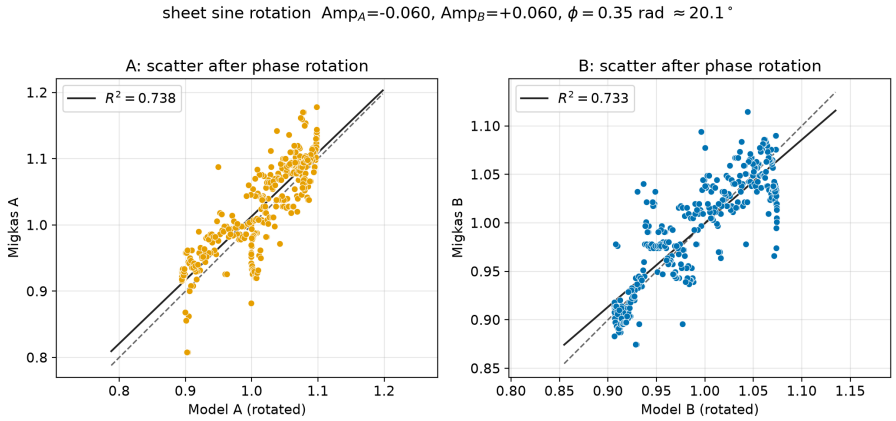


Figure 17: Migkas versus model scatter at Amp= -0.06 (sheet product).

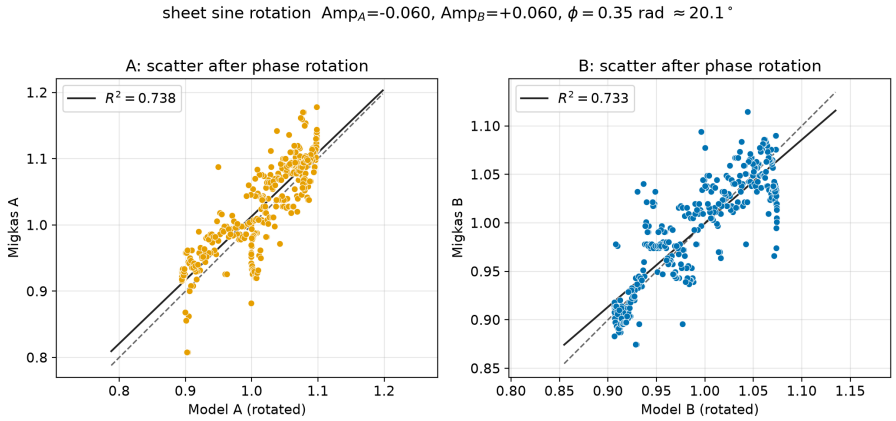


Figure 18: Migkas bars versus Amp$\times$ sin rotation model (scatter).

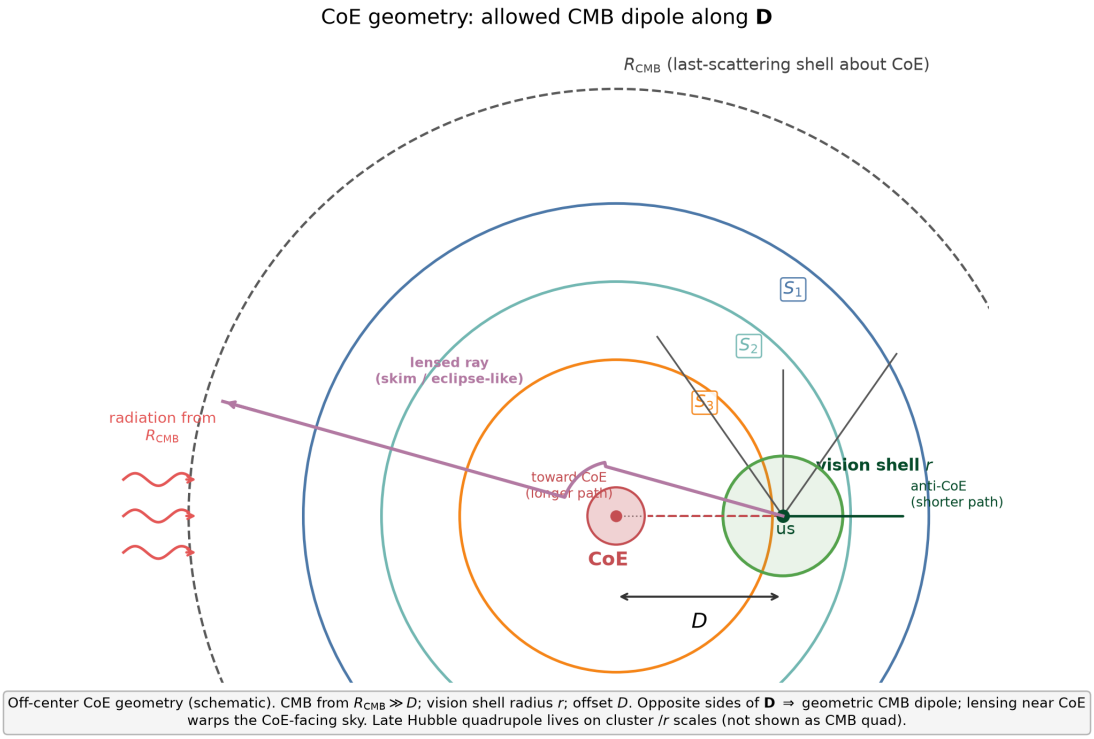


Figure 19: Schematic: off-center observer inside concentric shells; outer-shell heating offset can imprint a dipole.

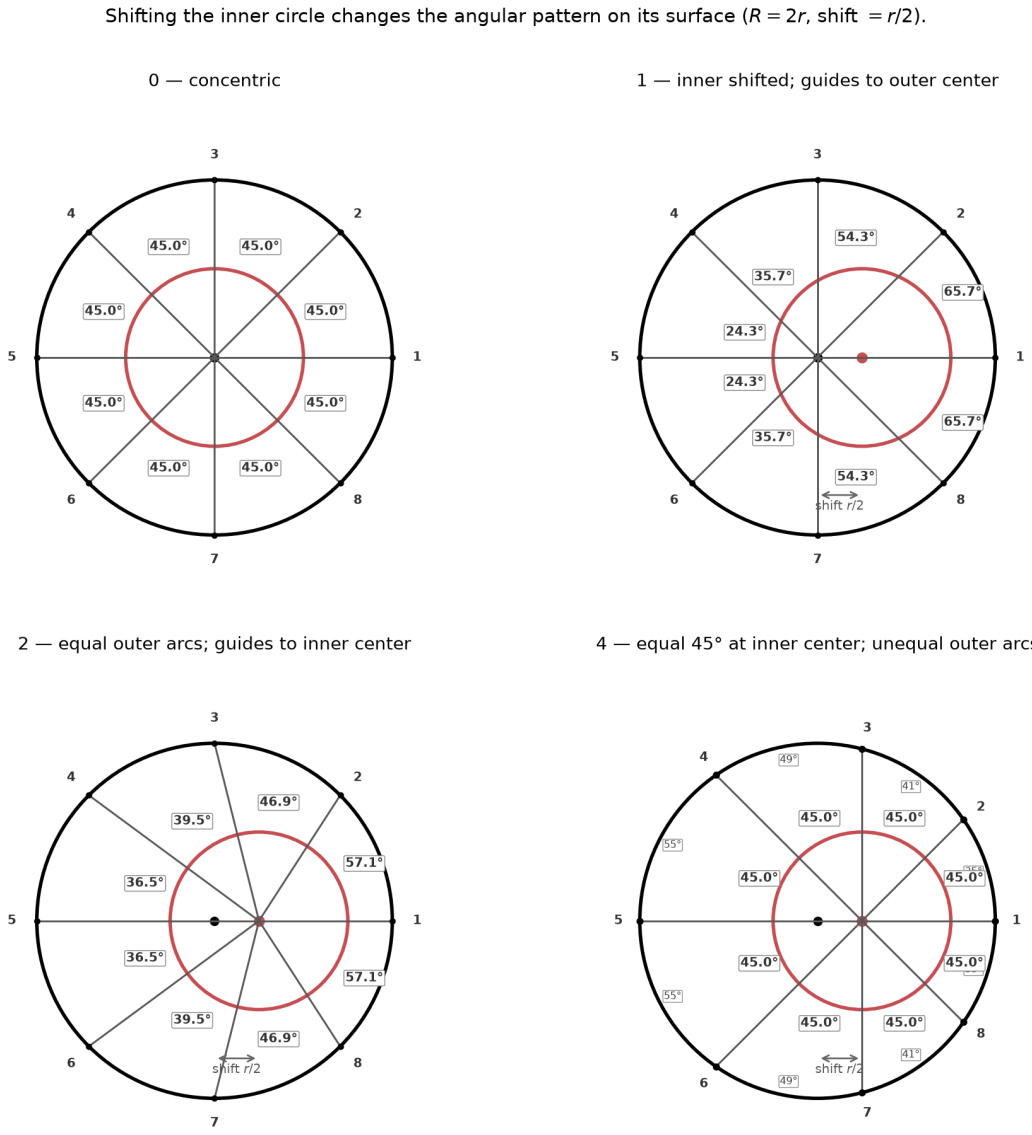


Figure 20: Four shell-heating scenarios overview. Scenario 1 is dropped for large shift; Scenarios 0, 2, 4 are the working set.

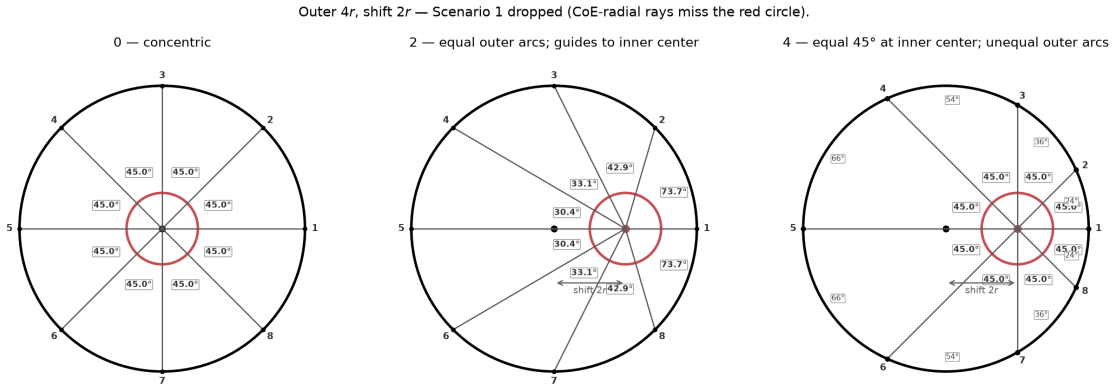


Figure 21: Scenarios 0, 2, 4 at illustrative $R = 4r$, shift $2r$ (geometry demo; CMB amplitude matching uses $R/D \sim 203$).

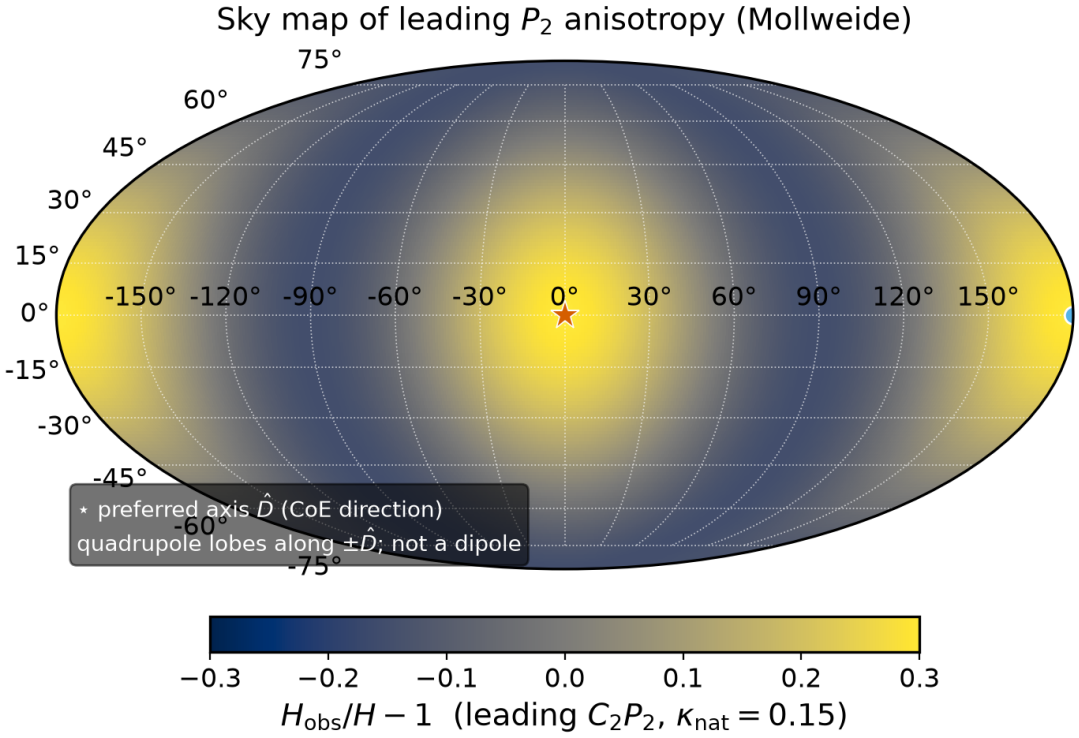


Figure 22: Existing program Mollweide (`sky_mollweide.png`). Provisional; included honestly as a stage-setting sky product.

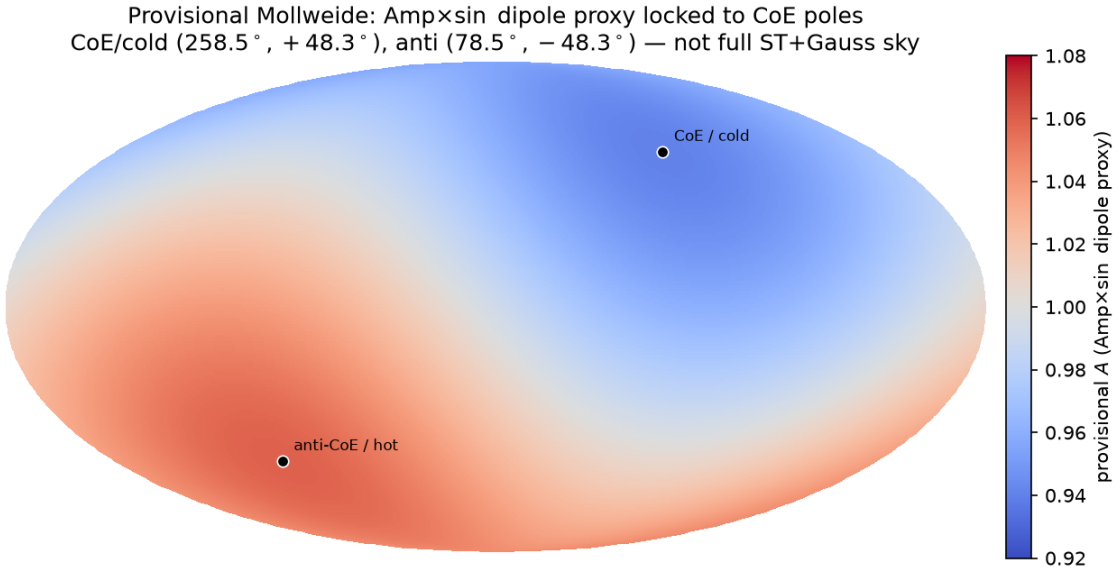


Figure 23: Provisional Mollweide of an Amp× sin dipole proxy with CoE poles locked to $(258.5°, +48.3°)$ / $(78.5°, -48.3°)$.

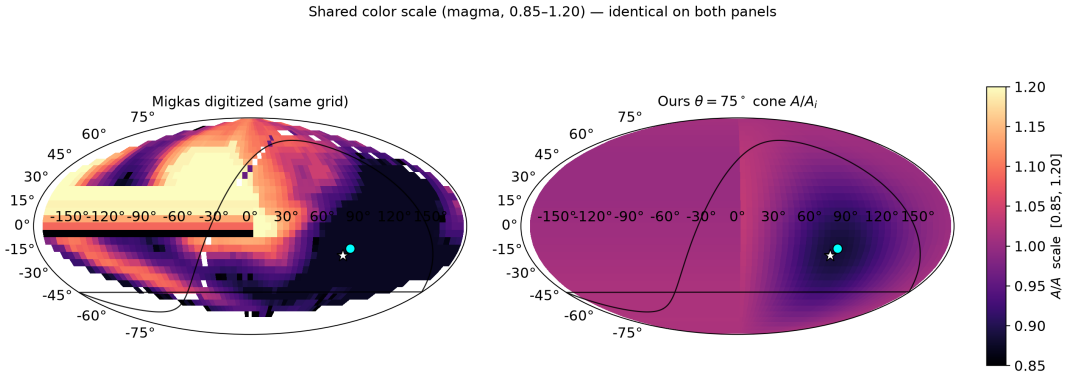


Figure 24: Side-by-side shared-scale Mollweide comparison product from `map_compare/` (program archive).

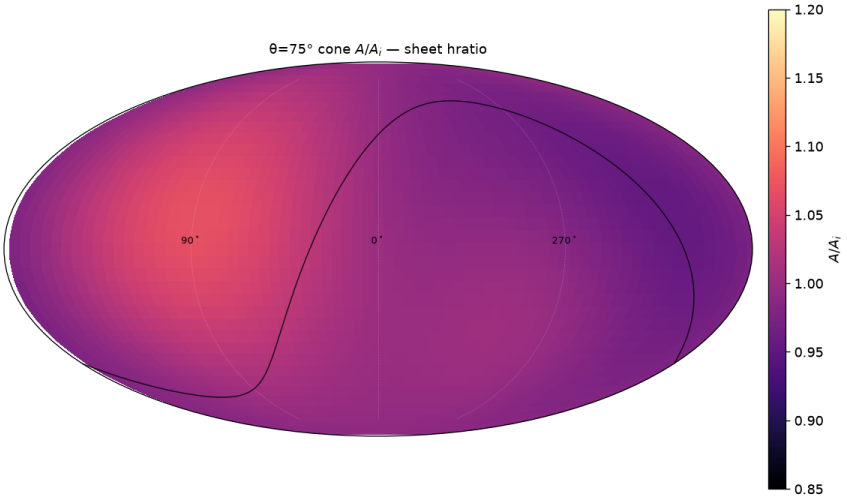


Figure 25: Example CARD sky map: $\theta = 75°$ cone, sheet H-ratio (from `map_compare/cone_thetas/cards/`).

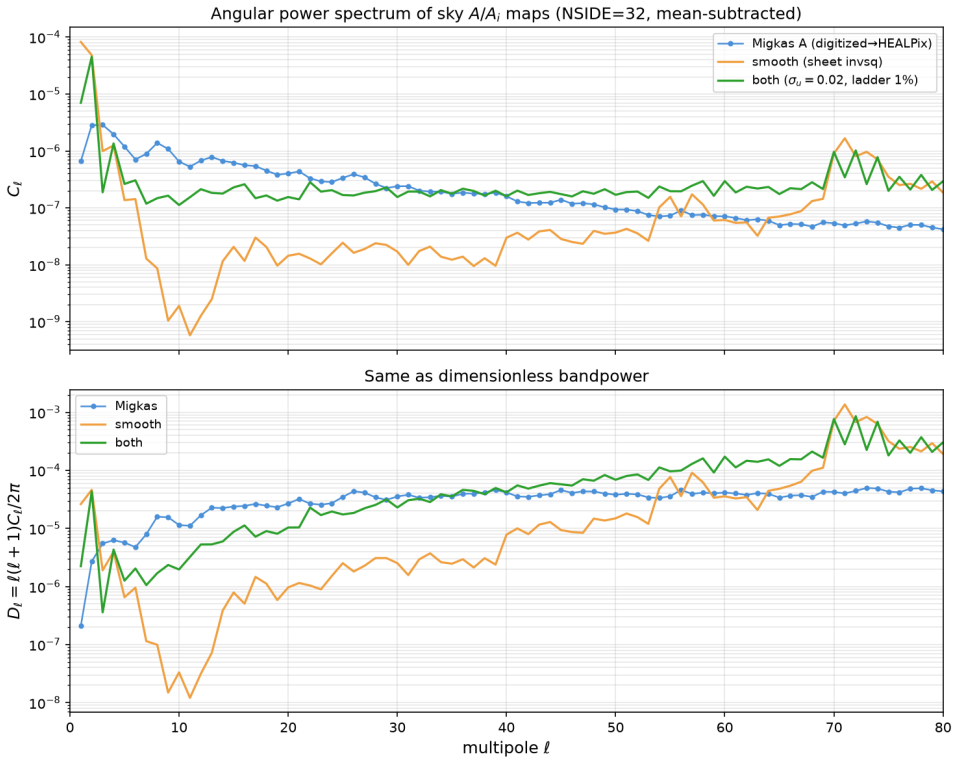


Figure 26: C_ℓ / smoothed A map product associated with Migkas-smooth comparisons (`Cl_A_migkas_smooth_both.png`).